

Unit 13, Lesson 4: Solving for Unknown Angles in Triangles

Benchmark

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

Learning Target

- ☐ I can write and solve algebraic equations to solve mathematical problems involving angle relationships of triangles.

13.4.1 Warm-Up: Riding on Square Wheels

The National Museum of Mathematics has an exhibition called the Square-Wheel Trike. The exhibition is pictured below with two children riding on the square-wheeled tricycles. The exhibit's description says, "Take a smooth ride on square wheels."



1. Describe any angles or shapes observed in the exhibition.
2. Why do you think it's possible for the ride to be smooth on square wheels?

13.4.2 Collaboration: Find that Angle

1. Use the words and angle measures from the bank to complete the table below.

adjacent

common

complementary

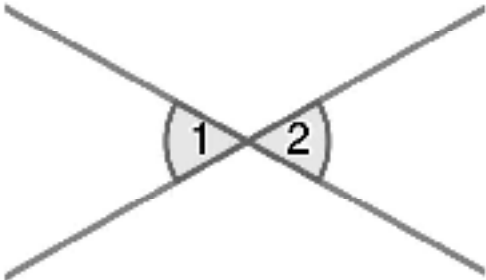
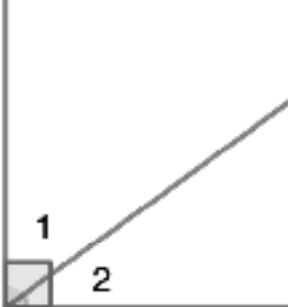
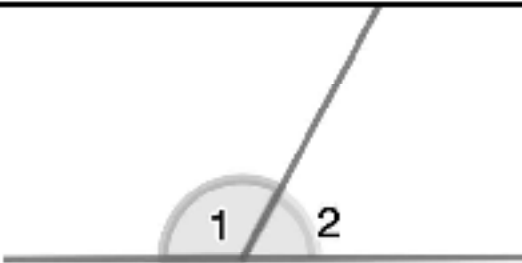
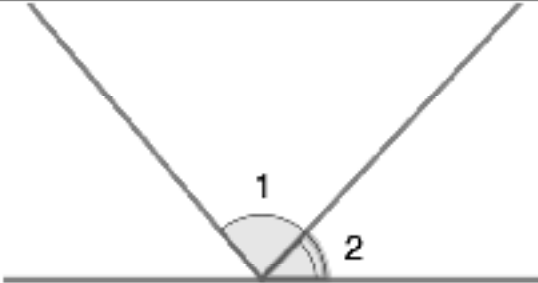
90°

equal

supplementary

vertical

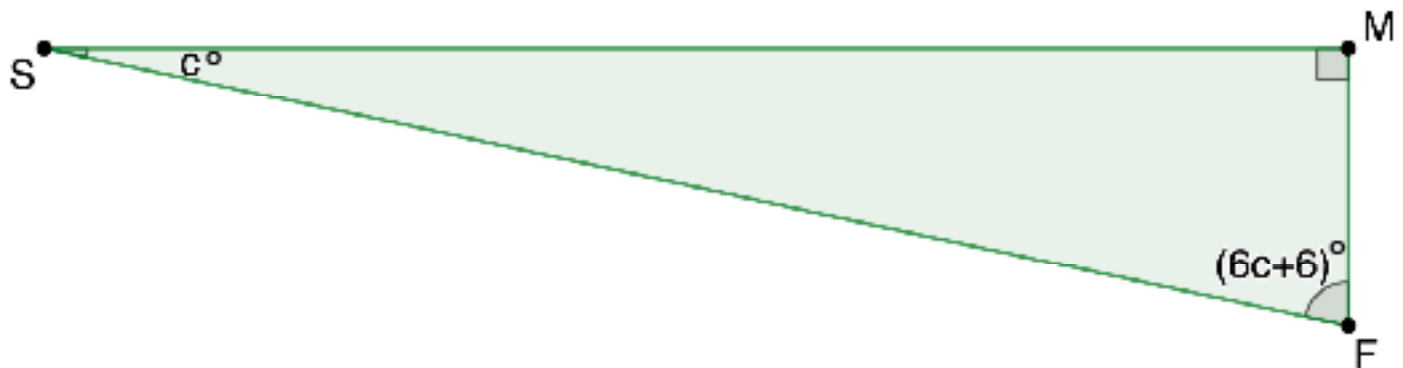
180°

Image	Angle Relationship	Description
	_____ angles	$\angle 1$ and $\angle 2$ have _____ measures.
	_____ angles	$\angle 1$ and $\angle 2$ sum to _____.
	_____ angles	$\angle 1$ and $\angle 2$ sum to _____.
	_____ angles	$\angle 1$ and $\angle 2$ have a _____ vertex and side.

13.4.3 Guided Instruction: Writing Equations to Find Missing Angles in Triangles

Algebraic equations can be written to express the relationship between the interior and exterior angles of triangles. The equations can then be solved to find the measures of unknown angles.

1. Triangle SMF is shown.



- a. Complete the statement.

The relationship of the

☐ exterior
☐ interior

 angles of the triangle can be used to find the measures of the missing angles.

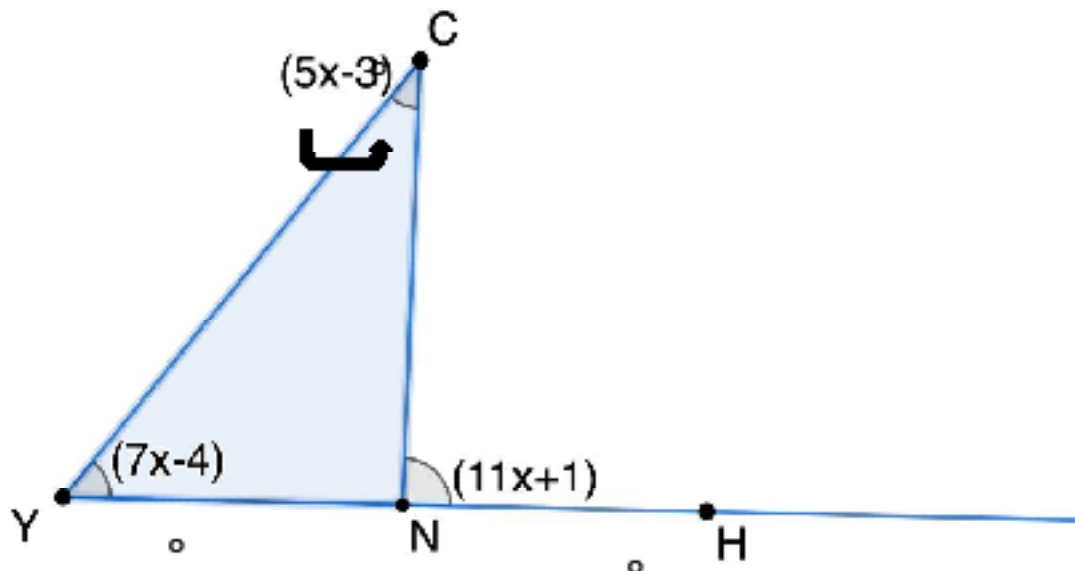
- b. Write an equation that can be used to find the value of c .

- c. Solve the equation.

- d. Use the solution to find the measure of each angle.

Angle	Measure
$\angle FSM$	
$\angle SFM$	

2. Triangle CYN is shown.



- a. Complete the statement.

The relationship between the
☐ exterior
☐ interior
 angles of the triangle can be used to find the measures of the missing angles.

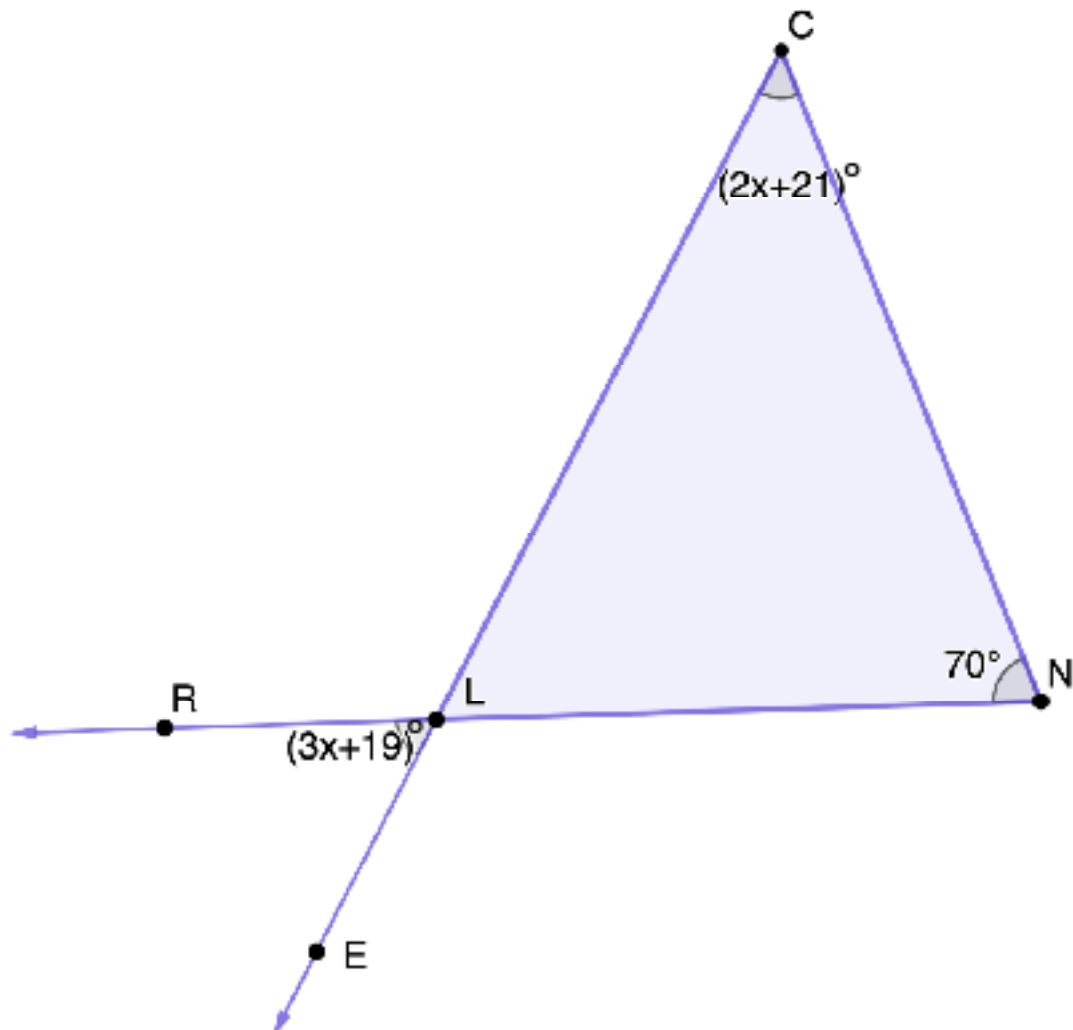
- b. Write an equation that can be used to find the value of x .

- c. **Solve the equation.**
-
-
-
-
-
-
-
-
-
-
- d. **Use the solution to find the measure of each angle.**

Angle	Measure
$\angle NCY$	
$\angle CYN$	
$\angle CNH$	

13.4.4 Collaboration: Using Multiple-Angle Relationships to Find Missing Angles

1. Use the figure shown to complete the following with your partner.



- a. Complete the table by writing the angle relationship for each set of angles using one of the phrases from the bank below. Then, complete the description of the relationship between the angles.

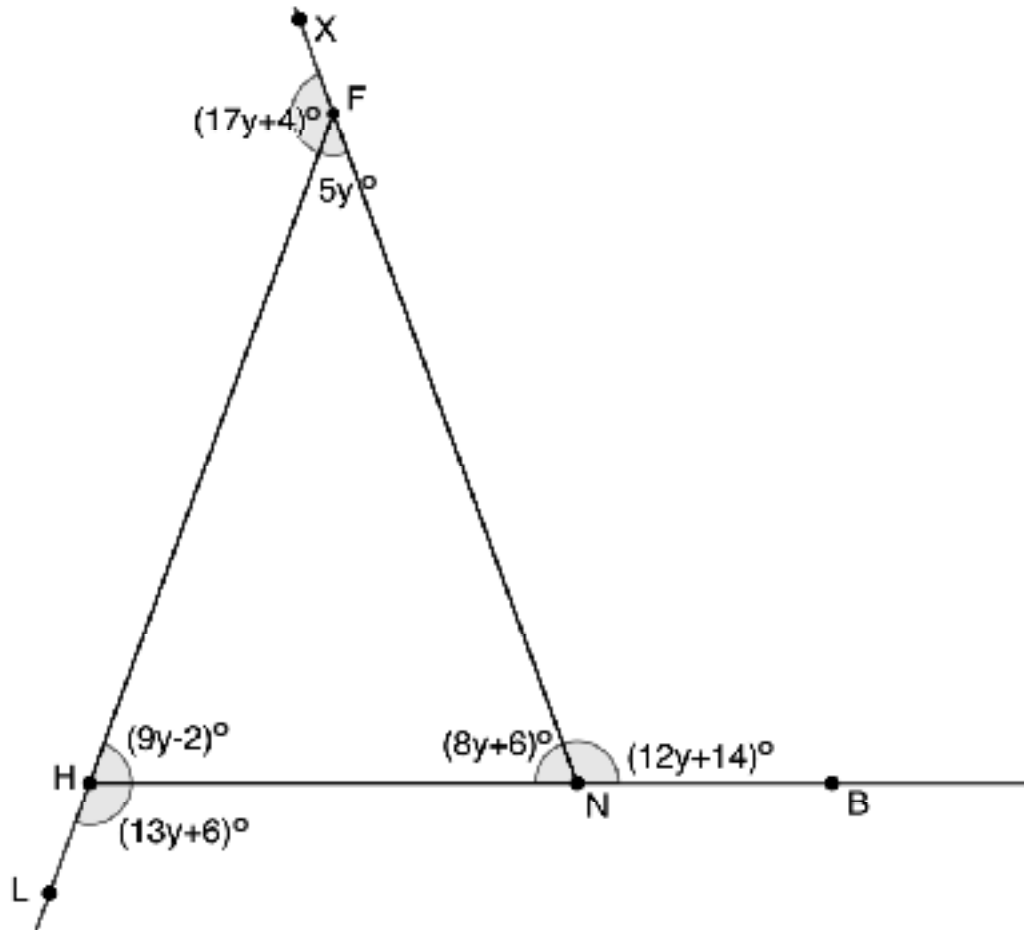
exterior angles to $\angle CLN$	supplementary angles
interior angles of triangle CLN	vertical angles

Angles	Angle Relationship	Description of Relationship
$\angle RLE$ and $\angle CLN$		Angles measures are _____.
$\angle RLE$ and $\angle RLC$		Angles sum to _____.
$\angle RLC$ or $\angle ELN$		_____ to the sum of $\angle NCL$ and $\angle LNC$.
$\angle CLN$, $\angle LNC$, and $\angle NCL$		Angles sum to _____.

- b. Different combinations of the relationships described in the table above can be used to find the value of x . Discuss with your partner which relationships you prefer to use to find the value of x .
- c. Write and solve an equation to find the value of x .
- d. Find the measure of $\angle CLN$.

$$m\angle CLN = \underline{\hspace{1cm}}^\circ$$

2. A figure including triangle FHN is shown.



- a. The three given equations can be used to find the value of y using the angle relationships in the figure. Identify which angle relationship was used to write each equation.

Equation	Angle Relationship
$13y + 6 + 9y - 2 = 180$	☐ exterior angle of a triangle ☐ interior angles of a triangle ☐ supplementary angles
$12y + 14 = 9y - 2 + 5y$	☐ exterior angle of a triangle ☐ interior angles of a triangle ☐ supplementary angles
$5y + 9y - 2 + 8y + 6 = 180$	☐ exterior angle of a triangle ☐ interior angles of a triangle ☐ supplementary angles

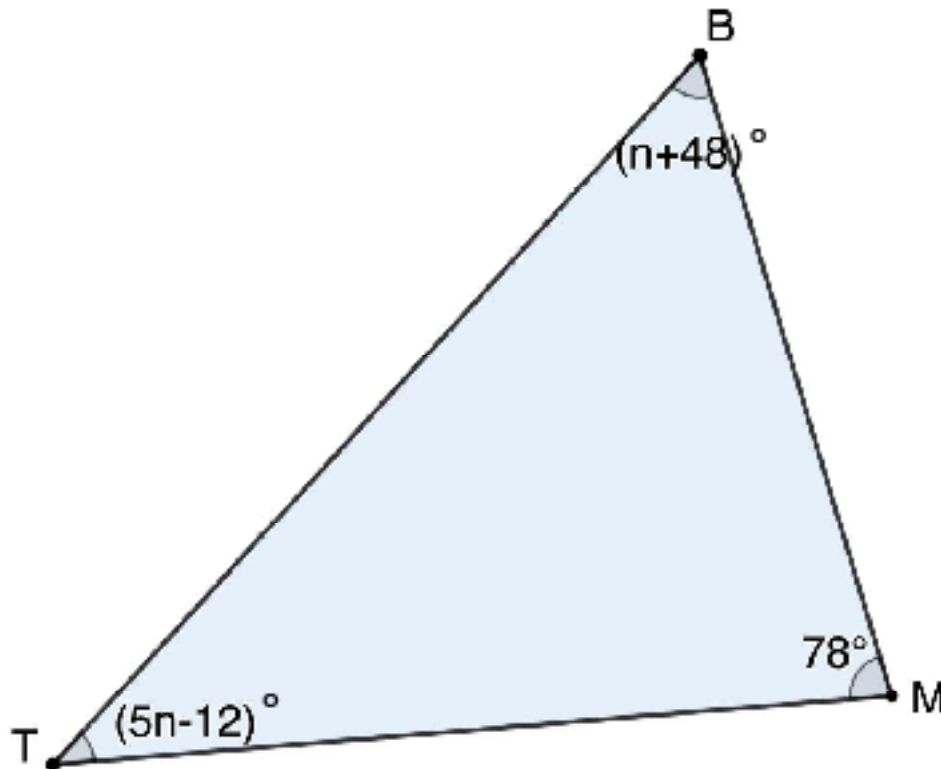
b. Choose one of the equations from Part A and use it to find the value of y .

c. Find the value of each angle measure in the table.

Angle	Measure
$\angle HFX$	
$\angle HNF$	
$\angle NHL$	

13.4.5 Your Turn!

1. Triangle BTM is shown.

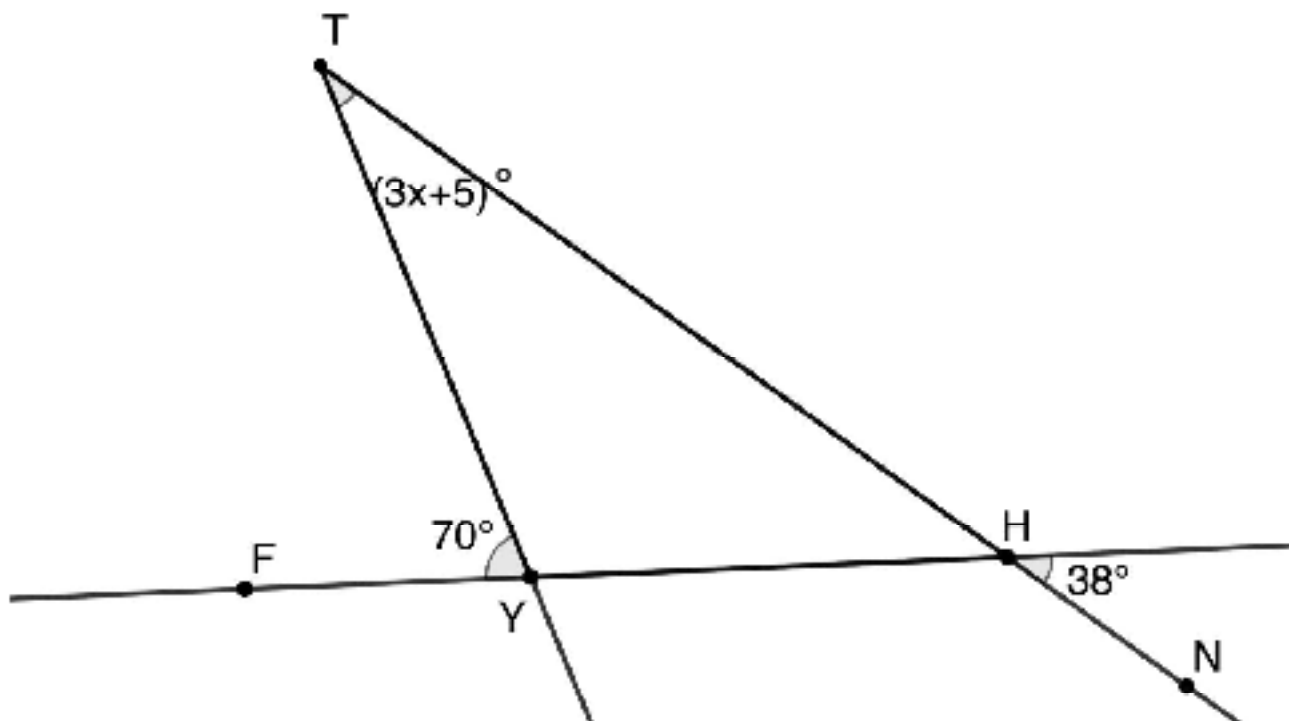


- a. Write and solve an equation to find the value of n .
- b. Find the measure of angle BTM .

$$m\angle BTM = \underline{\hspace{1cm}}^\circ$$

13.4.6 Wrap-Up: Ticket Out the Door

1. Use the figure shown to complete the following.



- a. Write and solve an equation to find the value of x .

- b. Find the measure of $\angle HTY$.

$$m\angle HTY = \underline{\hspace{1cm}}^\circ$$